

Pricing Freddie Mac STACR 2026-DNA1 credit risk transfer notes

Deal structure, scenarios, a loan-level default and prepayment model, the waterfall, prices, and a check against the market for the A-1, M-1 and M-2 notes

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Five parts, one pipeline from loan tape to price

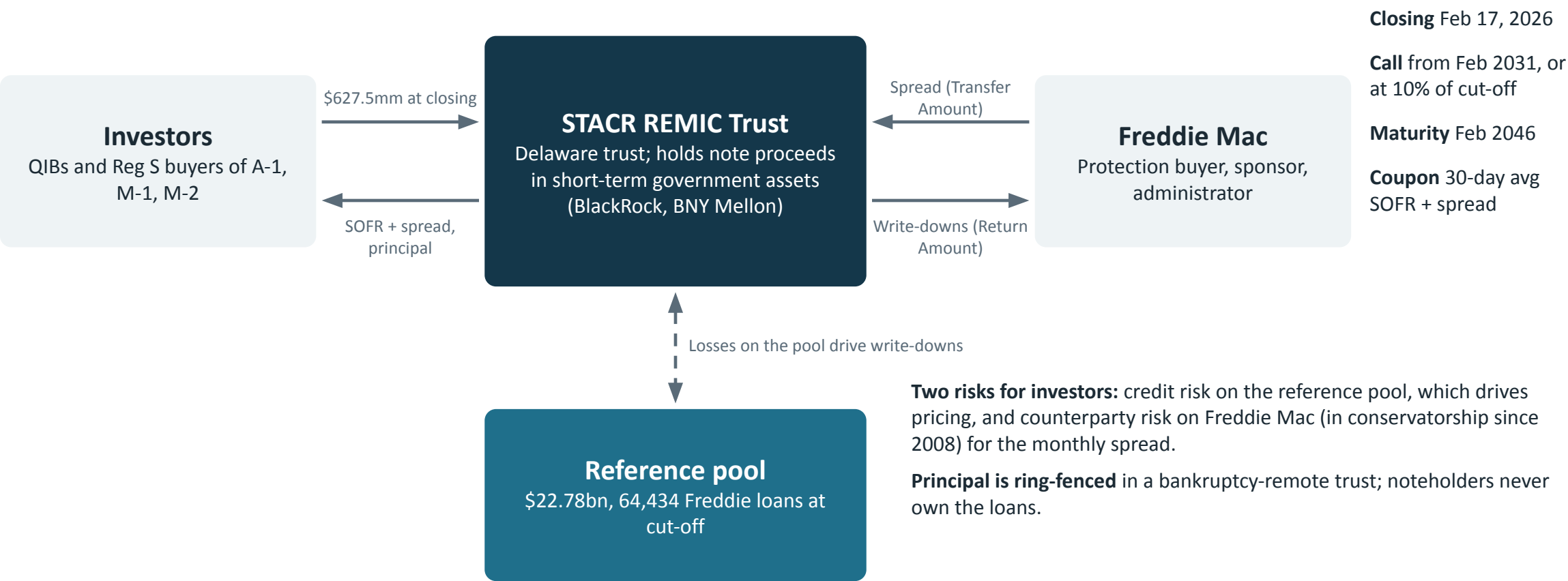
#	Part	Presenter	Time
1	Deal structure, loss and principal rules	Al Yazid Bensaid	3.5 min
2	Loan data and cleaning	Haocheng Sun	1.5 min
3	Rate and house-price scenarios	Coco Ma	2.5 min
4	Default and prepayment model	Reza Zamani	4 min
5	Waterfall and pricing	Smarajit Paul Choudhury	4 min
2	Check against Bloomberg	Haocheng Sun	1.5 min
3	Verdicts and recommendations	Coco Ma	1.5 min

About 20 minutes, then questions

What we set out to answer

- How is the deal organized, and what protects investors?
- How fast does the pool prepay and how much does it lose?
- Do those losses reach A-1, M-1 or M-2? If not, what is the risk?
- What are the notes' lives and prices across scenarios?
- Do our lives and prices agree with Bloomberg?
- Which investors should own which class?

Investors buy notes from a trust, not mortgages



The tranche stack: offered notes span 1.90–4.80%

Class	Original (\$mm)	Attach	Detach	Coupon	Rating (S&P / DBRS)	Holder
A-H	21,687.7	4.800%	100.000%	n/a	n/a	Freddie Mac
A-1	275.9	3.525%	4.800%	SOFR + 0.85%	A+ / A (low)	Investors
M-1	275.9	2.250%	3.525%	SOFR + 1.00%	BBB+ / BBB (high)	Investors
M-2A	37.85	2.075%	2.250%	SOFR + 1.30%	BBB+ / BBB	Investors
M-2B	37.85	1.900%	2.075%	SOFR + 1.30%	BBB+ / BBB (low)	Investors
B-1H	102.5	1.450%	1.900%	n/a	n/a	Freddie Mac
B-2H	273.4	0.250%	1.450%	n/a	n/a	Freddie Mac
B-3H	57.0	0.000%	0.250%	n/a	n/a	Freddie Mac

Each offered class has an "H" twin (about 5% of the band) that Freddie Mac keeps and that shares losses and principal pro rata.
Source: PPM Table 3.

After the Sep 2026 payment

A-1 **\$203.5mm** factor 0.7375

M-1 **\$157.8mm** factor 0.5720

M-2A **\$37.85mm** factor 1.0000

M-2B **\$37.85mm** factor 1.0000

September factors validated against Bloomberg; each H twin uses its class's factor and A-H is the residual. Pool: \$19.25bn.

Losses go bottom up; principal splits in two buckets

Losses: bottom up

- Actual loss: credit-event UPB + missed interest + prior forgiveness – net liquidation proceeds (incl. MI)
- Write-down order: B-3H, B-2H, B-1H, M-2B, M-2A, M-1, A-1
- Modification losses cut interest first, then principal
- Recoveries write classes back up, top down

Principal: two buckets

- Senior share = senior balances / pool; 96.475% at closing
- A-1 fast-pay: 3.75% of its band a month in months 1–12, then 1.50% to month 36
- Subordinate bucket pays M-1, then M-2A, then M-2B (sequential)
- If any trigger fails, all principal goes senior

Triggers for M-1 and M-2 (all must pass)

Test	Passes if
Minimum credit enhancement	Subordinate share $\geq$ 3.525%
Cumulative net loss	Loss / cut-off $\leq$ 0.10% in year 1, +0.10% a year to 1.30%
Delinquency	6-month avg distressed balance < 50% of subordinate cushion

- The minimum CE test passes with no cushion at closing, so any loss reaching the B tranches switches principal to senior
- A-1 loses its fast-pay schedule for good if cumulative loss ever exceeds 1.00%

Starting point: the September 2026 loan tape

56,433

loans on the Sep 2026
tape (64,434 at cut-off)

\$19.25bn

pool after Sep 2026
(\$22.78bn at cut-off)

6.76%

WA coupon vs 7.28%
PMMS today

759

WA FICO

70.5

WA current LTV (75.6 at
origination; no MI)

0.79%

of balance 30+ days
late; DTI 38.5

1 Load

Newest dated tape: Freddie
Mac disclosure in the
Bloomberg layout, matched to
Bloomberg CLP

2 Clean

Standardize field names, units
and dates; balance-weighted
pool statistics; written to
data/processed

3 Class factors

September factors for every
class, checked against
Bloomberg; A-H is the residual

4 Hand-off

Loans already 60+ days late
(about \$52mm) start in the
liquidation pipeline

Two methods: simulation and historical stress replays

20,000

simulated paths

53

monthly steps to the Feb 2031 call

319

months of calibration data
(2000–2026)

5

factors moved jointly

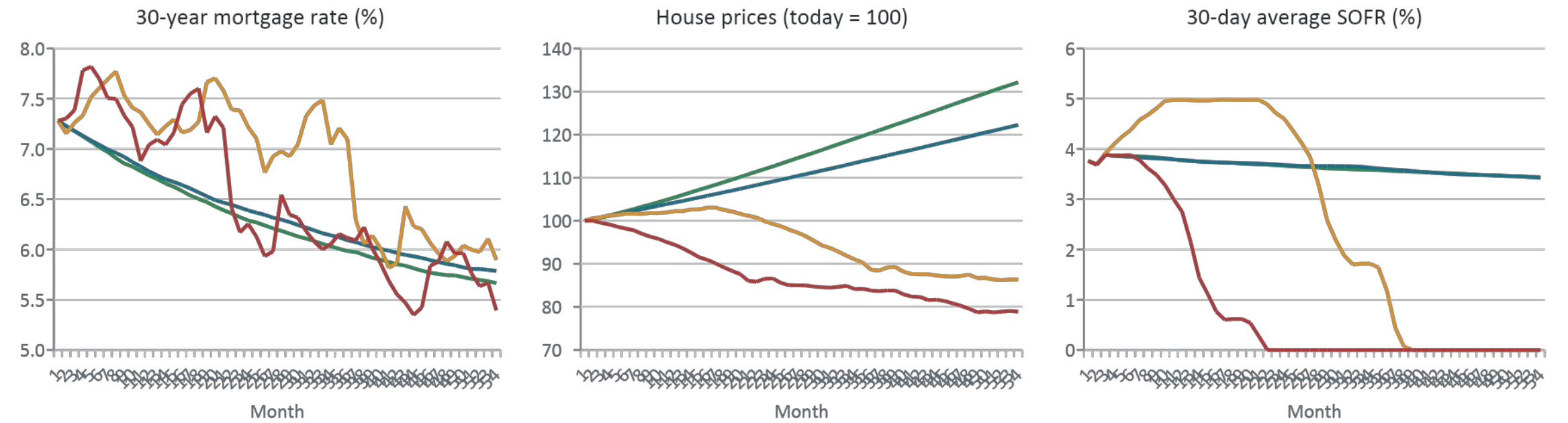
Good and base: simulation

- AR(1) for each factor: short rate, 2y and 10y Treasury, mortgage spread, HPI growth
- Residuals bootstrapped in 6-month blocks, all factors from the same months, so co-movement is kept
- Paths ranked by house-price outcome: good = 75–85th percentile, base = 45–55th; each scenario is the band's average path

Moderate and severe: history

- Replay 53-month windows of joint rate and HPI changes from 2000–2026 (266 candidate windows)
- Moderate: 10th-percentile HPI window, Dec 2005 – May 2010
- Severe: worst window, Mar 2007 – Aug 2011
- Changes re-anchored to today's rates

All paths start at today's 7.28% rate; house prices split



Scenario	Method	HPI change by call	Mortgage rate at call	SOFR at call
Good	Simulation, 75–85th pct	+32.1%	5.67%	3.42%
Base	Simulation, 45–55th pct	+22.3%	5.79%	3.41%
Moderate	Replay Dec 2005 – May 2010	–13.7%	5.90%	0.00%
Severe	Replay Mar 2007 – Aug 2011	–21.1%	5.39%	0.00%

Three behaviors, applied loan by loan each month

Prepayment

$$\text{CPR} = \text{ramp} \times [\text{turnover} + \text{refi max} \times S(\text{coupon} - \text{PMMS}) - \text{burnout}]$$

- Logistic S-curve in the refinancing incentive
- Fitted: turnover 5.5%, refi add-on up to 41.4%, midpoint 0.71 pp, ramp about 6 months
- CPR turned into a monthly SMM

Default

Poisson GLM for the monthly default rate

- Mark-to-market LTV (steeper above 80) moves with the scenario's house prices
- FICO, DTI, investor status, recent 30-day delinquency, pre-2009 vintage
- Loans 60+ days late on the tape start in the pipeline

Liquidation and loss

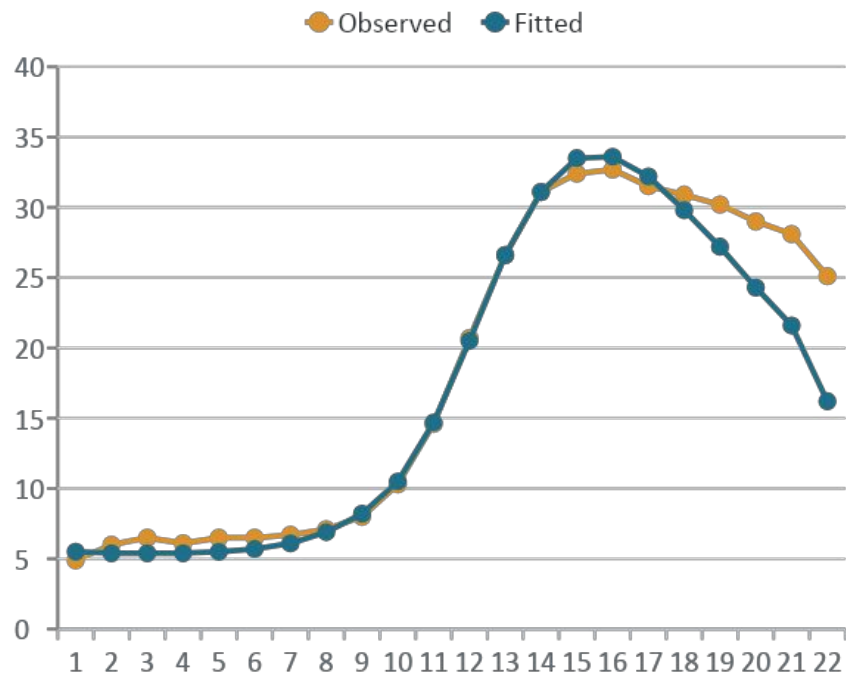
$$\text{Loss} = \text{UPB} + \text{missed interest} + \text{costs} - \text{sale proceeds} - \text{MI}$$

- About 19 months from default to credit event; no cures
- Sale at a fitted 48% distressed discount to HPI value
- 57% of distressed loans modified (0.57 pp rate cut)

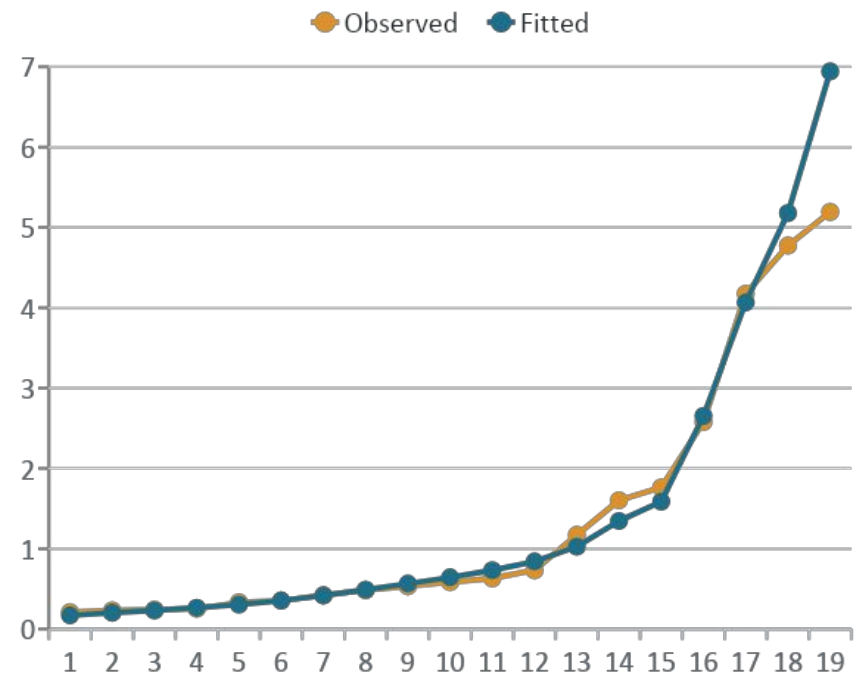
Each month defaults, then scheduled amortization, then prepayments. Output per scenario, months 1–53: scheduled principal, prepayments, credit events, losses, modification losses, distressed balance.

Calibrated on 48 million Freddie Mac loan-months

CPR (%) by refi incentive (pp); $R^2 = 0.91$



Default rate (% a year) by mark-to-market LTV



- Prepayment: weighted least squares on observed speeds by incentive, age and burnout
- Default: Poisson GLM on 38M loan-months and 22,598 default spells, every coefficient of the expected sign

977,577

loans, 27 vintages
(2000–2026)

19 mo

median time to credit event

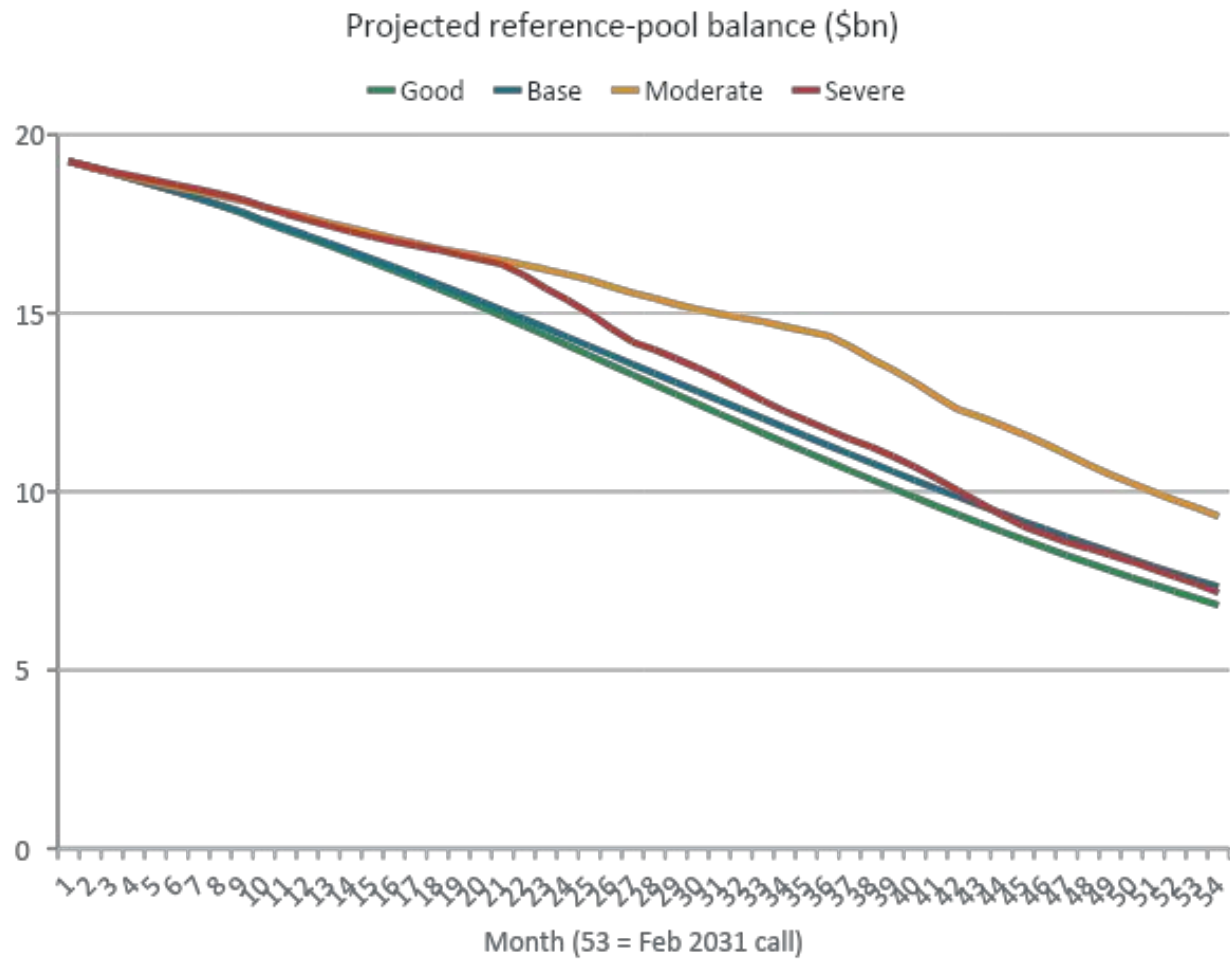
48%

distressed-sale discount

45.5%

model severity vs 44.4%
observed

The pool pays down slowly until rates fall below its coupon



Scenario	Year-1 CPR	Month 24	At the call	Half paid by
Good	10.9%	\$13.8bn	\$6.83bn	month 40
Base	10.6%	\$14.1bn	\$7.33bn	month 42
Moderate	7.6%	\$15.9bn	\$9.30bn	month 52
Severe	8.0%	\$15.0bn	\$7.17bn	month 43

Slow start. At 7.28% the pool is out of the money, so year-1 CPR is only 8–11%.

Then fast. Once rates fall below the coupon, CPR reaches about 24–31% by the call.

Most extension: moderate. 48% of today's pool is still outstanding at the call.

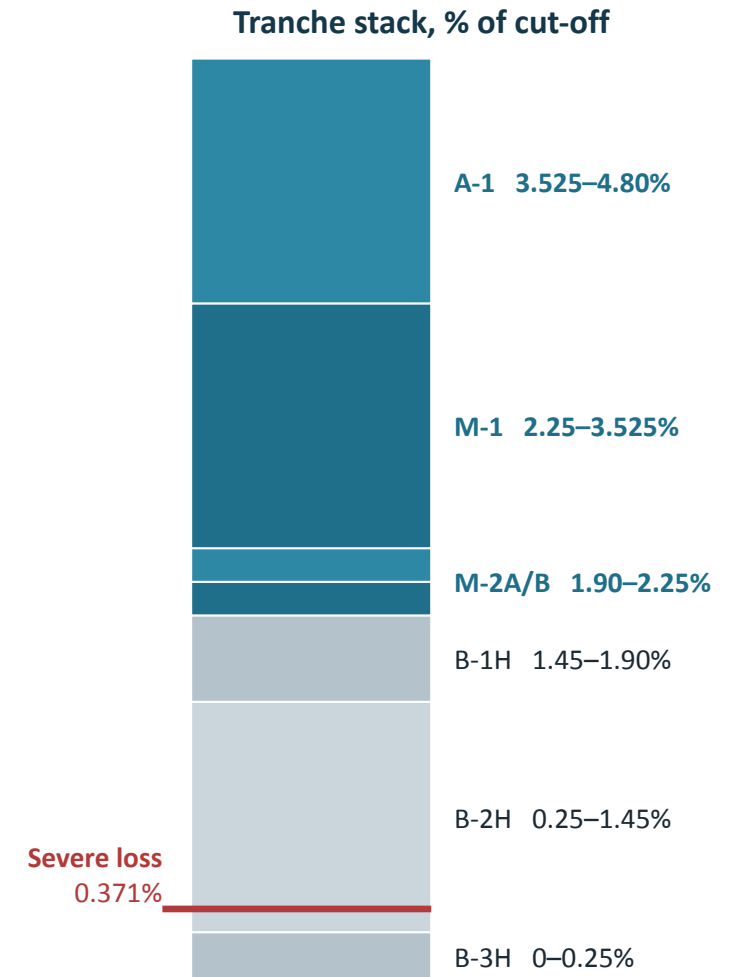
Pool losses of 0.20–0.37% never reach A-1, M-1 or M-2

	Good	Base	Moderate	Severe
HPI change to 2031	+32%	+22%	−14%	−21%
Credit events	\$121.8mm	\$124.0mm	\$136.0mm	\$149.3mm
Liquidation losses	\$46.4mm	\$50.3mm	\$68.6mm	\$84.6mm
Loss, % of cut-off	0.204%	0.221%	0.301%	0.371%
Average severity	38.1%	40.6%	50.4%	56.7%

Good and base stay inside B-3H; moderate and severe run into B-2H. Freddie Mac keeps both.

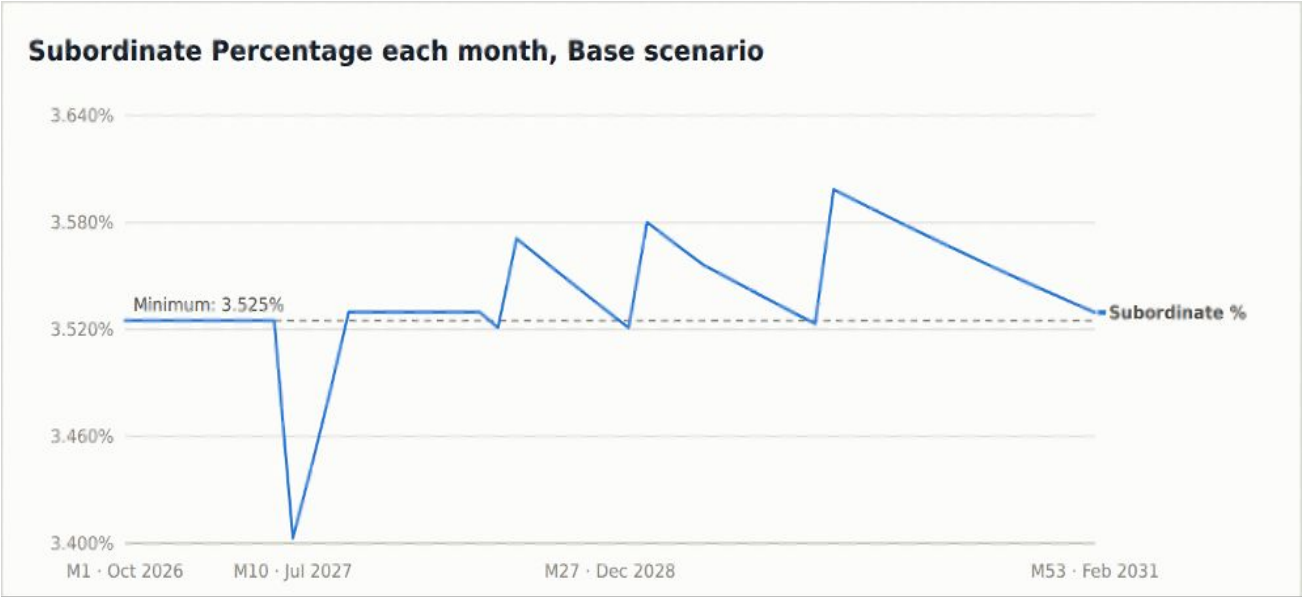
M-2B attaches at 1.90%, about 5 times the severe-case loss. The worst case uses about a fifth of the cushion.

So the question for investors is not whether they lose principal but when they get it back.



Offered notes in blue; grey is retained by Freddie Mac

The deal sits on its trigger, so losses keep cutting off M-note principal



Scenario	Months failing (of 53)	Initial lockout
Good	6	Months 10–12
Base	6	Months 10–12
Moderate	11	Months 10–14
Severe	10	Months 10–13

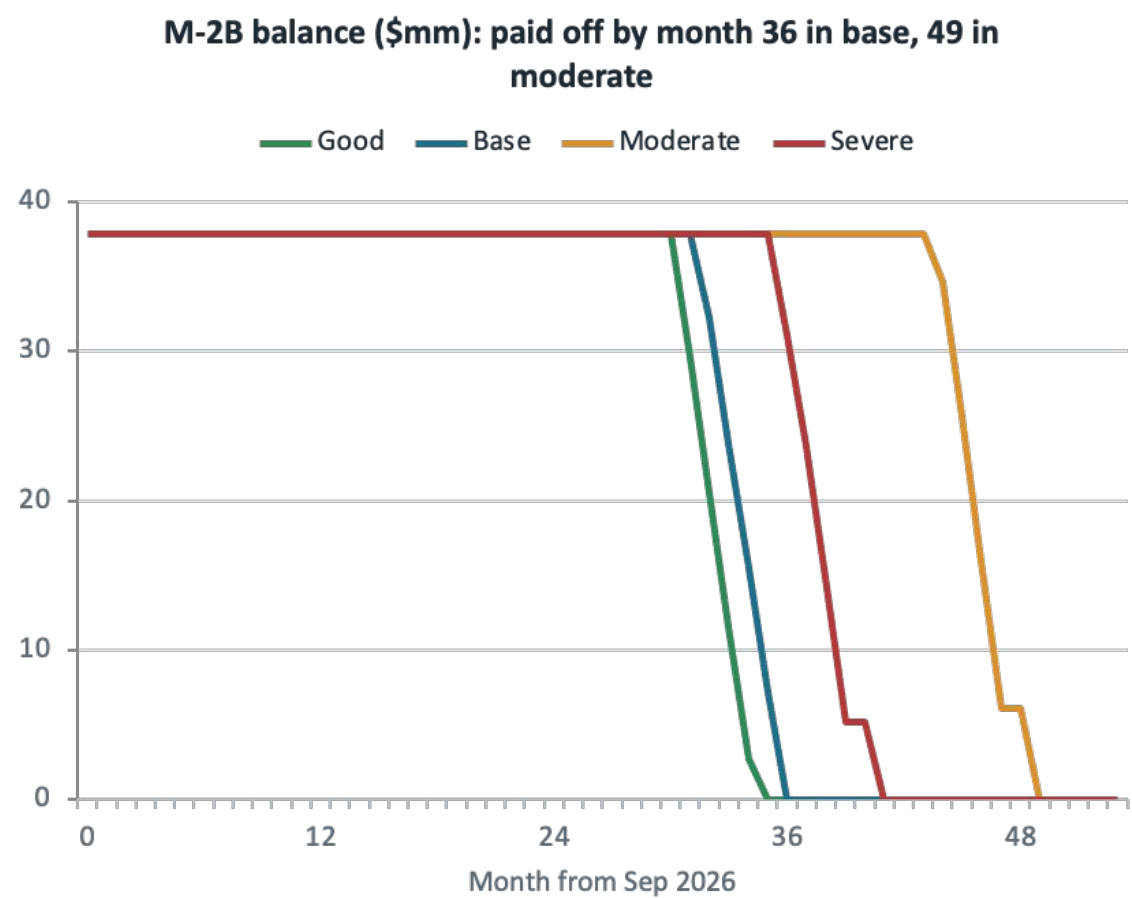
What happens

- **No cushion:** the test needs a Subordinate Percentage of at least 3.525%, and the deal starts exactly there.
- **Pro rata holds it flat:** while the test passes, the M notes get their pro-rata share of principal, so the share does not move.
- **Losses push it down:** each loss, even one Freddie Mac absorbs in B-3H, lowers the share. Once it dips below 3.525%, all principal goes senior until it climbs back.
- **Month 9 is the biggest:** the tape's delinquent loans liquidate together: a \$23.31mm loss and a three-month lockout. One-date liquidation overstates it.

The cumulative-loss and delinquency tests pass in every month; only minimum CE fails. Months counted to the assumed Feb 2031 call. Chart: base scenario.

For investors: M-note principal can stop for months at a time even when Freddie Mac absorbs the loss, so credit events show up as timing risk.

No write-downs; the risk is extension in the moderate case



For investors: A-1 behaves like a 1.4-year cash instrument; an M-2B buyer should plan to hold for up to 4 years.

WAL, yrs	Good	Base	Moderate	Severe	PPM
A-1	1.40	1.40	1.40	1.40	1.59
M-1	1.16	1.20	1.87	1.54	1.75
M-2A	2.28	2.43	3.47	2.74	4.11
M-2B	2.70	2.82	3.84	3.16	4.79
Write-down	0	0	0	0	

PPM: expected WAL at issue (Feb 2026), 10% CPR, no losses, called Feb 2031. Ours run from Sep 2026, so they are shorter by construction.

- A-1 is locked in:** it takes only its fixed schedule until payment 37 (month 30), when it is retired: 1.40 years in every scenario.
- M notes extend about a year in moderate**, where rates stay above the pool's 6.76% coupon for three years. M-2B's last payment moves from month 35 (good) to 49.
- Severe extends less** because rates fall and refinancing picks up.

Each 100 bp of spread moves a note by about its life in points

Base case	WAL (yrs)	DM 0 bp	DM 200 bp	Points per 100 bp
A-1 (85 bp)	1.40	101.26	98.57	1.35
M-1 (100 bp)	1.20	101.28	98.95	1.17
M-2A (130 bp)	2.43	103.16	98.51	2.33
M-2B (130 bp)	2.82	103.63	98.27	2.68

M-2B by scenario	WAL (yrs)	Points per 100 bp	Cost of +50 bp
Good	2.70	2.58	-1.27
Base	2.82	2.68	-1.32
Moderate	3.84	3.55	-1.74
Severe	3.16	3.06	-1.50

Full prices per 100 face incl. about 0.10 accrued since Sep 25, 2026; discounted at each scenario's SOFR path plus the DM. Points per 100 bp = (price at DM 0 – price at DM 200) / 2; cost of +50 bp is from the coupon spread.

Reading the prices

- Points $\approx$ life.** Each 100 bp of DM moves a note's price by roughly its WAL in points, so longer notes swing more.
- M-2B is most exposed:** 2.7 points per 100 bp in base and 3.6 in moderate, where its life stretches to about 3.8 years.
- Engine check:** at its own coupon spread every class prices at 100.10 (par plus accrued) in every scenario, as a floater should.
- DM from a price:** we solve for the DM by bisection, which is how we compare with Bloomberg next.

For investors: a 50 bp spread widening costs M-2B about 1.3 points in base and 1.7 in moderate; A-1 about 0.7.

Bloomberg agrees on credit; M-2 is priced for extension

Class	Factor	BBG WAL	Our base WAL (range)	BBG bid / ask	Ask DM	Coupon	Our base DM at ask	Our moderate DM
A-1	0.7375	1.39	1.40 (1.40–1.40)	99.844 / 99.978	87	85	87	87
M-1	0.5720	1.23	1.20 (1.16–1.87)	99.875 / 100.027	98	100	98	99
M-2A	1.0000	2.78	2.43 (2.28–3.47)	99.377 / 99.720	141	130	142	139
M-2B	1.0000	3.48	2.82 (2.70–3.84)	98.761 / 99.148	157	130	162	154

Bloomberg as of October 2, 2026; DMs in bp. "Our DM at ask" is the DM that makes our projected cash flows worth Bloomberg's ask price plus accrued.

Balances match

Bloomberg factors and balances equal the ones our waterfall uses for every class. 16 output tests pass.

Timing is in range

Every Bloomberg WAL is inside our scenario range. A-1 and M-1 match our base; M-2 is near our moderate case.

Extension, not credit

M-2A and M-2B trade 11–27 bp over their 130 bp coupon. No scenario writes them down, so the extra spread is for slower paydown.

M-2B looks fair

At the ask, M-2B earns 162 bp on our base cash flows and 154 bp on moderate, against 157 bp quoted.

All five hypotheses hold, with a timing gap for M-2

#	We expected	We found
H1	A strong pool loses well under 1% of cut-off, even in a 2008-style fall	Supported: losses of 0.20–0.37% across scenarios
H2	The 1.90% below M-2B absorbs every projected loss	Supported: zero write-downs for every offered class in every scenario
H3	Out-of-the-money loans prepay slowly, so extension is the main risk	Supported: year-1 CPR of 7.6–10.9%; M lives extend about a year in moderate
H4	Floaters stay near par at their own spread	Supported: about 100.10 (par plus accrued) for every class
H5	Our timing and prices agree with the market	Supported, with a gap: A-1 and M-1 match our base; the market prices M-2 near our moderate case

Who should own which class

A-1

1.40-year life in every scenario, 85 bp

Banks and money-market-style buyers who want a SOFR floater with little extension

M-1

1.2 years, up to 1.9 in moderate; 100 bp

Insurers and asset managers who want extra spread with no projected loss

M-2A / M-2B

2.4–2.8 years, up to 3.8; 141–157 bp at the ask

Total-return credit funds and hedge funds that can hold through slower paydown

Limitations

- Starting delinquent loans liquidate on one date, which exaggerates the month-9 trigger failure
- No cures; national, not regional, house prices
- One Bloomberg snapshot (Oct 2, 2026), no price history
- Assumes the February 2031 call is exercised

All three classes: no projected loss. Investors also take Freddie Mac counterparty risk on the monthly spread.

Thank you. Questions?

Code and results:

github.com/rezazamani2329/Asset-Backed-Security-Market--UC-Berkeley-MFE-program